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Seat No:______

No. of Printed Pages: 02

SARDAR PATEL UNIVERSITY
S.Y.B.C.A - 4th Semester Examination -2023
US04CBCA54 || Operating Systems

Date : 8/8/2023
Time : 10:00 a.m. to 1:00 p.m.

Marks: 70

Q.1 Multiple Choice Questions **[10]**

1 Multi-tasking system also known as __________.
(a) Real-time (b) Time-sharing (c) Multi-user (d) None of these

2 A Program in execution is known as ________.
(a) Process (b) Files (c) Job (d) None of these

3 A Lazy Swapper is also known as________.
(a) Pages (b) Pager (c) Frames (d) None of these

4 Swapping of process requires
(a) Backing Store (b) Main Memory (c) Operating System (d) All of these

5 A_______ process produces information that is consumed by a consumer process.
(a) Consumer (b) Computation (c) Producer (d) None

6 The situation when waiting process never change its state because of no availability of resources is called ___________.
(a) DeadLock (b) DeadState (c) DeadQueue (d) None

7 Which command is used to remove directories?
(a) deldir (b) rm (c) rmdir (d) dirdel

8 ________command is used to get information about currently logged in user on to system.
(a) who (b) login (c) syslog (d) imlogin

9 For copying files and directories in linux __________ command is used.
(a) cat (b) copy (c) cpy (d) cp

10 First UNIX was developed by a ____________.
(a) Ken Thompson (b) Linus Torvald (c) Linux Torvald (d) None

Q.2 Short Question (Any TEN) **[20]**

1 Define Operating system and list out types of OS.
2 Draw the diagram of PCB.
3 What is the solution of starvation? How is it work?
4 Explain Internal Fragmentation in short.
5 Calculate Page faults using FIFO algorithm for following reference string:
(Number of Frames = 3)
Reference string = 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1
6 What is Compaction? What is the use of it?
7 Why is LINUX an open source?
8 List out features of Linux.
9 Justify “Linux is a Secure Operating System”.
10 Write a brief note on who command.
11 Discuss mkdir and rmdir in brief.
12 What is shell script ?

[P.T.O]

Q.3

[A] Explain Round-Robin scheduling algorithm with example [5]

[B] Explain process state and its transitions... [5]

OR

Q.3

[A] List out various types of OS. Explain Time Sharing Operating System. [5]

[B] Define OS. Explain Services of Operating System. [5]

Q. 4

[A] Explain Demand Paging. [4]

[B] What is Memory Management? Explain Memory Management Allocation Techniques [6]

OR

Q.4

[A] Explain Swapping. [4]

[B] List out different page Replacement algorithms and Explain Optimal Page Replacement Algorithm. [6]

Q.5

[A] Explain Process Synchronization. [4]

[B] What do you mean by Deadlock? Explain all necessary conditions for occurrence of deadlock. [6]

OR

Q.5

[A] Explain algorithm 3 for two-process solution. [4]

[B] List out differences between cooperative and independent process. Also explain advantages of cooperative process. [6]

Q.6 Explain ls, dir, cd commands with use, syntax and example [10]

OR

Q.6 Discuss all looping statements of Linux in detail with example. [10]

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